VIDEO GAME CREATION

Lesson 4 - Programming Basics III: Direct Hit

Semester 1, Week 4

- **Objectives:** Explain what collision/overlap detection is and why every engine needs an explicit way to ask "did these two things touch"; explain the difference between a trigger/area-based check and a solid physical collision; explain why responding to one collision by destroying two objects is safe even in edge cases (near-simultaneous hits); add a working single-enemy-plus-collision mechanic to the ongoing Semester 1 project.
- **IGDA Tie-In:** Programming - collision/physics systems, event-driven programming (introductory)
- **Featured Engine (lecture demo):** Godot Engine
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 3 projects (already saved from last week)

Agenda

0:00-0:05	Welcome back, Lesson 3 recap, quick share of any working shooting mechanics
0:05-0:25	Collision Detection: How Engines Know Things Touched (live demo in Godot)
0:25-0:45	Cleanup Revisited: Destroying Two Things at Once, Safely
0:45-0:50	Placement: Things You Spawn vs. Things You Place
0:50-1:00	Introduce lab exercise, hand out engine-specific handouts
1:00-1:50	LAB: Add an enemy and collision detection to your ongoing [YourName]Game project
1:50-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): A single enemy sits on screen; a bullet touching it destroys both.
- Level 2 (Complete): The enemy is destroyed cleanly and exactly once, with no errors even under repeated or near-simultaneous hits; misses still behave exactly like last week.
- Level 3 (Refined): Add a hit sound, a distinct enemy look, and/or clean hit-response structure.

See: Lesson 4 Teacher's Script, Lesson 4 Prep Checklist, and the per-engine Student Handouts.